

Dummit & Foote Notes

by: blanketism

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0 Preliminaries

0.1 Basics

◆ **Order/Cardinality:** denoted as $|A|$ for a set A . If A is finite, then $|A|$ is the number of elements in A .

◆ **Cartesian Product:** The Cartesian product of two sets A and B is the collection of elements

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

◆ **Function:** Denoted as $f : A \rightarrow B$ for sets A and B . The value of f at a is denoted as $f(a)$ for all $a \in A$. Moreover, to denote $f(a) = b$, we write $f : a \mapsto b$, or $a \mapsto b$.

◆ **Domain:** The set A in the function $f : A \rightarrow B$.

◆ **Codomain:** The set B in the function $f : A \rightarrow B$.

◆ **Range/Image:** The subset of B defined as

$$f(A) = \{b \in B \mid f(a) = b \text{ for some } a \in A\}.$$

◆ **Preimage/Inverse Image:** For $C \subseteq B$, the subset of A given by

$$f^{-1}(C) = \{a \in A \mid f(a) \in C\}.$$

◆ **Fiber:** For $b \in B$, the preimage of b , i.e., $f^{-1}(b)$.

◆ **Identity Map:** The function on a set A defined as $\text{id}_A : A \rightarrow A$ where $\text{id}_A(a) = a$ for all $a \in A$.

◆ **Composition Map:** Given functions $f : A \rightarrow B$ and $g : B \rightarrow C$, it is the map $g \circ f : A \rightarrow C$ is defined as

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A.$$

Let $f : A \rightarrow B$ be a function.

◆ **Injective/Injection:** for all $a_1, a_2 \in A$, $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.

◆ **Surjective/Surjection:** for all $b \in B$, there exists $a \in A$ such that $f(a) = b$, i.e., $f(A) = B$.

◆ **Bijective/Bijection:** f is both injective and surjective.

◆ **Left Inverse:** A function $g : B \rightarrow A$ is a left inverse of $f : A \rightarrow B$ if $g \circ f = \text{id}_A$.

◆ **Right Inverse:** A function $h : B \rightarrow A$ is a right inverse of $f : A \rightarrow B$ if $f \circ h = \text{id}_B$.

♣ Proposition 0.1 ► Characterizations of Injectivity, Surjectivity, and Bijectivity

Let $f : A \rightarrow B$ be a function.

- (1) f is injective if and only if f has a left inverse.
- (2) f is surjective if and only if f has a right inverse.
- (3) f is a bijection if and only if there exists $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$.
- (4) If A and B are finite sets with $|A| = |B|$, then the following are equivalent:
 - (a) f is bijective.
 - (b) f is injective.
 - (c) f is surjective.

Proof.

- (1) $\Rightarrow$ Suppose f has a left inverse $g : B \rightarrow A$. Then for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$, then

$$a_1 = g(f(a_1)) = g(f(a_2)) = a_2,$$

so f is injective.

◀ Suppose f is injective. Then for each $b \in f(A)$, there exists a unique $a \in A$ such that $f(a) = b$. Now, fix some $a_0 \in A$. Then we can define a function $g : B \rightarrow A$ as

$$g(b) = \begin{cases} a & \text{if } b \in f(A) \text{ and } f(a) = b, \\ a_0 & \text{if } b \in B \setminus f(A). \end{cases}$$

To see g is well-defined, note that if $b \in f(A)$, then by injectivity of f , there is a unique $a \in A$ such that $f(a) = b$. Then for all $a \in A$,

$$g(f(a)) = a,$$

so $g \circ f = \text{id}_A$ and g is a left inverse of f .

- (2) ➡ Suppose f has a right inverse $h : B \rightarrow A$. Then for all $b \in B$, we may set $a = h(b) \in A$. Then $f(a) = f(h(b)) = b$, so f is surjective.

◀ Suppose f is surjective. Then $f^{-1}(b)$ is nonempty for all $b \in B$. Define a function $h : B \rightarrow A$ by choosing some $a \in f^{-1}(b)$ for each $b \in B$, where this choice is possible by the axiom of choice. Then for all $b \in B$,

$$f(h(b)) = b,$$

so $f \circ h = \text{id}_B$ and h is a right inverse of f .

- (3) ➡ Suppose f is bijective. Then by (1) and (2), f has a left inverse $g : B \rightarrow A$ and a right inverse $h : B \rightarrow A$. Then for all $b \in B$,

$$g(b) = g(f(h(b))) = h(b),$$

where $g(f(h(b))) = g(b)$ because $f(h(b)) = b$ from h being a right inverse of f , and $g(f(h(b))) = h(b)$ because g is a left inverse of f . Thus, $g = h$ and $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$.

◀ Suppose there exists $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$. Then by (1) and (2), f is injective and surjective, so f is bijective.

- (4) • If f is bijective, then by (1) and (2), f is injective and surjective.
 • If f is injective, then distinct elements of A map to distinct elements of B . Since $|A| = |B|$, this implies that f is surjective, so f is bijective.
 • If f is surjective, then all elements of B are mapped to by elements of A . Since $|A| = |B|$, this implies that f is injective, so f is bijective. $\square$

◆ **Inverse:** The function $g : B \rightarrow A$ in [Proposition 0.1](#) (3), denoted as $f^{-1} : B \rightarrow A$.

◆ **Permutation:** A bijection $f : A \rightarrow A$.

◆ **Restriction:** Let $A \subseteq B$ and $f : B \rightarrow C$ be a function. The **restriction** of f to A is the function $f|_A : A \rightarrow C$ defined as

$$f|_A(a) = f(a) \quad \text{for all } a \in A.$$

◆ **Extension:** Let $A \subseteq B$ and $g : A \rightarrow C$ be a function. The **extension** of g to B is a function $f : B \rightarrow C$ such that $f|_A = g$. This function need not be unique, and may not even exist.

Let A be a set.

- ◆ **Binary Relation:** A subset $R \subseteq A \times A$, where we write $a \sim b$ if $(a, b) \in R$.
- ◆ **Reflexive:** A relation R is reflexive if $a \sim a$ for all $a \in A$.
- ◆ **Symmetric:** A relation R is symmetric if $a \sim b$ implies $b \sim a$ for all $a, b \in A$.
- ◆ **Transitive:** A relation R is transitive if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.
- ◆ **Equivalence Relation:** A relation R that is reflexive, symmetric, and transitive.
- ◆ **Equivalence Class:** For $a \in A$, the equivalence class of a is the set

$$[a] = \{b \in A \mid b \sim a\}.$$

Elements of the $[a]$ are said to be **equivalent** to a and are **representatives** of $[a]$.

- ◆ **Partition:** Let I be an indexing set. A partition of A is a collection of nonempty subsets $\{A_i\}_{i \in I}$ such that

- (1) $A = \bigcup_{i \in I} A_i$.
- (2) $A_i \cap A_j = \emptyset$ for all $i, j \in I$ with $i \neq j$.

♣ Proposition 0.2 ► Fundamental Theorem of Equivalence Relations

Let A be a nonempty set.

- (1) If $\sim$ is an equivalence relation on A , then the set of equivalence classes of $\sim$ form a partition of A .
- (2) Let I be an indexing set. If $\{A_i \mid i \in I\}$ is a partition of A , then there is an equivalence relation on A whose equivalence classes are the sets A_i for $i \in I$.

Proof.

- (1) Let $\sim$ be an equivalence relation on A , and put

$$B = \bigcup_{a \in A} [a]$$

to be the union of all equivalence classes of $\sim$. For any $a \in A$, we have $a \sim a$ by reflexivity, so $a \in [a]$ and thus $a \in B$. Hence, $A \subseteq B$. Conversely, let $b \in B$. Then $b \in [a]$ for some $a \in A$, so $b \sim a$ and thus $b \in A$. Hence, $B \subseteq A$ and thus $A = B$.

Now, let $a, b \in A$. If $[a] \cap [b] = \emptyset$, then we are done. Otherwise, let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$, so by symmetry and transitivity, we have $a \sim c$ and $c \sim b$, which implies $a \sim b$. Now, let $x \in [a]$. Then $x \sim a$ and $a \sim b$, so by transitivity, we have $x \sim b$ and thus $x \in [b]$. Hence, $[a] \subseteq [b]$. A similar argument shows that $[b] \subseteq [a]$, so $[a] = [b]$, and thus the equivalence classes of $\sim$ form a partition of A .

- (2) Let $\{A_i \mid i \in I\}$ be a partition of A . Define a relation $\sim$ on A by $a \sim b$ if and only if there exists $i \in I$ such that $a, b \in A_i$. Then $\sim$ is reflexive because for any $a \in A$, there exists $i \in I$ such that $a \in A_i$, so $a \sim a$. Moreover, $\sim$ is symmetric because if $a \sim b$, then there exists $i \in I$ such that $a, b \in A_i$, so $b \sim a$. Finally, $\sim$ is transitive because if $a \sim b$ and $b \sim c$, then there exist $i, j \in I$ such that $a, b \in A_i$ and $b, c \in A_j$. Since $b \in A_i \cap A_j$, we must have $i = j$ because the A_i are disjoint for $i \neq j$. Hence, $a, c \in A_i$, so $a \sim c$. Therefore, $\sim$ is an equivalence relation, and its equivalence classes are precisely the sets A_i for $i \in I$. $\square$

0.2 Properties of the Integers

◆ **Well-Ordering of $\mathbb{Z}^+$:** If $A \subseteq \mathbb{Z}^+$ is nonempty, there exists $a' \in A$ such that $a' \leq a$ for all a in A , called the **minimal element** of A .

◆ **Divides:** For $a, b \in \mathbb{Z}$ with $a \neq 0$, we say that a **divides** b , denoted $a \mid b$, if there exists $c \in \mathbb{Z}$ such that $b = ac$. Moreover, if a does not divide b , we write $a \nmid b$.

◆ **Greatest Common Divisor:** Denoted as (a, b) for $a, b \in \mathbb{Z}$ with $a \neq 0$, it is the unique $d \in \mathbb{Z}^+$ such that

- $d \mid a$ and $d \mid b$, and
- if $c \in \mathbb{Z}$ such that $c \mid a$ and $c \mid b$, then $c \mid d$.

◆ **Relatively Prime:** Nonzero $a, b \in \mathbb{Z}$ such that $(a, b) = 1$.

◆ **Least Common Multiple:** Denoted as $\text{lcm}(a, b)$ for $a, b \in \mathbb{Z}$ with $a \neq 0$, it is the unique $\ell \in \mathbb{Z}^+$ such that

- $a \mid \ell$ and $b \mid \ell$, and
- if $c \in \mathbb{Z}$ such that $a \mid c$ and $b \mid c$, then $\ell \mid c$.

Note

For $a, b \in \mathbb{Z}$ with $a \neq 0$, we have

$$(a, b) \cdot \text{lcm}(a, b) = |ab|.$$

◆ **The Division Algorithm:** For $a, b \in \mathbb{Z}$ with $b > 0$, there exist unique $q, r \in \mathbb{Z}$ such that

$$a = bq + r \quad \text{and} \quad 0 \leq r < b.$$

We say that q is the **quotient** and r is the **remainder** of the division of a by b .

◆ **The Euclidean Algorithm:** For $a, b \in \mathbb{Z}$ with $b > 0$, we can find (a, b) by repeatedly applying the division algorithm to a and b until we reach a remainder of 0. More precisely, we can find (a, b) by finding a sequence of integers $r_0, r_1, \dots, r_n$ such that

$$\begin{aligned} a &= bq_0 + r_0, \\ b &= r_0q_1 + r_1, \\ r_0 &= r_1q_2 + r_2, \\ &\vdots \\ r_{n-2} &= r_{n-1}q_n + r_n, \\ r_{n-1} &= r_nq_{n+1} + 0. \end{aligned}$$

The last nonzero remainder r_n is (a, b) .

◆ **$\mathbb{Z}$ -Linear Combination:** For $a, b \in \mathbb{Z}$, an expression of the form $ax + by$ for some $x, y \in \mathbb{Z}$.

Note

We may reverse the Euclidean algorithm (also known as the extended Euclidean algorithm) to obtain the $\mathbb{Z}$ -linear combination of a, b that equates to (a, b) .

✓ **Example ▶ Calculating a Linear Combination**

Let $a = 63$ and $b = 39$. Using the Euclidean algorithm, we find

$$\begin{aligned} 63 &= 39(1) + 24, \\ 39 &= 24(1) + 15, \\ 24 &= 15(1) + 9, \\ 15 &= 9(1) + 6, \\ 9 &= 6(1) + 3, \\ 6 &= 3(2) + 0. \end{aligned}$$

Thus, $(63, 39) = 3$. Reversing the Euclidean algorithm, we find

$$\begin{aligned} 3 &= 9 - 6(1) = 9 - (15 - 9(1))(1) \\ &= 9(2) - 15(1) = (24 - 15(1))(2) - 15(1) \\ &= 24(2) - 15(3) = 24(2) - (39 - 24(1))(3) \\ &= 24(5) - 39(3) = (63 - 39(1))(5) - 39(3) \\ &= 63(5) - 39(8). \end{aligned}$$

- ◆ **Prime:** A nonzero integer p whose only positive divisors are 1 and p .
- ◆ **Composite:** A nonzero integer n that is not prime.

📌 **Note**

If a prime $p \mid ab$ for some $a, b \in \mathbb{Z}$, then either $p \mid a$ or $p \mid b$.

- ◆ **Fundamental Theorem of Arithmetic:** Every integer $n > 1$ can be written as a product of primes, and this factorization is unique up to the order of the factors.

📌 **Note**

If n has two prime factorizations

$$n = \prod_{i=1}^k p_i^{\alpha_i} = \prod_{j=1}^m q_j^{\beta_j}$$

for primes p_i, q_j and $\alpha_i, \beta_j \in \mathbb{Z}^+$, then $k = m$ and, after reordering, $p_i = q_i$ and $\alpha_i = \beta_i$ for all i .

📌 **Note**

Let $a, b \in \mathbb{Z}^+$ have prime factorizations

$$a = \prod_{i=1}^k p_i^{\alpha_i}, \quad b = \prod_{i=1}^k p_i^{\beta_i}$$

where p_i are distinct primes and $\alpha_i, \beta_i \geq 0$. Then

$$(a, b) = \prod_{i=1}^k p_i^{\min(\alpha_i, \beta_i)}, \quad \text{lcm}(a, b) = \prod_{i=1}^k p_i^{\max(\alpha_i, \beta_i)}.$$

- ◆ **Euler φ -Function:** For $n \in \mathbb{Z}^+$, $\varphi(n)$ is the number of integers k such that $1 \leq k \leq n$ and $(k, n) = 1$, along with two properties:

- For prime p and $\alpha \in \mathbb{Z}^+$, we have $\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$.
- For $m, n \in \mathbb{Z}^+$ such that $(m, n) = 1$, we have $\varphi(mn) = \varphi(m)\varphi(n)$.

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

Note

Let $n \in \mathbb{Z}^+$. Define the relation $\sim$ on $\mathbb{Z}$ by

$$a \sim b \iff n \mid (a - b).$$

- Since $a - a = 0$ and $n \mid 0$, we have $a \sim a$ for all $a \in \mathbb{Z}$.
- If $a \sim b$, then $n \mid (a - b)$, so $nk = a - b$ for some $k \in \mathbb{Z}$. Then $b - a = -nk$, so $n \mid (b - a)$, and thus $b \sim a$.
- If $a \sim b$ and $b \sim c$, then $n \mid (a - b)$ and $n \mid (b - c)$, so $nk_1 = a - b$ and $nk_2 = b - c$ for some $k_1, k_2 \in \mathbb{Z}$. Then

$$a - c = (a - b) + (b - c) = nk_1 + nk_2 = n(k_1 + k_2),$$

so $n \mid (a - c)$, and thus $a \sim c$.

◆ **Congruent Modulo n :** If $a \sim b$, we write $a \equiv b \pmod{n}$.

◆ **Congruence Class:** For $a \in \mathbb{Z}$, the **congruence class of a modulo n** is

$$\begin{aligned} \bar{a} &= \{b \in \mathbb{Z} \mid b \equiv a \pmod{n}\} \\ &= \{a + nk \mid k \in \mathbb{Z}\} \\ &= \{a, a \pm n, a \pm 2n, a \pm 3n, \dots\}. \end{aligned}$$

◆ **Integers Modulo n :** The set of all congruence classes modulo n is denoted by $\mathbb{Z}/n\mathbb{Z}$. There are precisely n distinct congruence classes modulo n , which can be represented by the integers $0, 1, 2, \dots, n-1$. Thus,

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}.$$

◆ **Modular Arithmetic:** For any $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$, we define

$$\bar{a} + \bar{b} = \overline{a + b} \quad \text{and} \quad \bar{a} \cdot \bar{b} = \overline{ab},$$

i.e., we add and multiply congruence classes by adding and multiplying their representatives, where any representative can be chosen.

Theorem 0.3 ► Modular Arithmetic is Well-Defined

The operations of addition and multiplication on $\mathbb{Z}/n\mathbb{Z}$ are well-defined, i.e., they do not depend on the choice of representatives. In other words, if $s_1, s_2 \in \mathbb{Z}$ and $t_1, t_2 \in \mathbb{Z}$ such that $\bar{s}_1 = \bar{s}_2$ and $\bar{t}_1 = \bar{t}_2$, then $\overline{s_1 + t_1} = \overline{s_2 + t_2}$ and $\overline{s_1 t_1} = \overline{s_2 t_2}$. More precisely, if

$$s_1 \equiv s_2 \pmod{n} \quad \text{and} \quad t_1 \equiv t_2 \pmod{n},$$

then

$$s_1 + t_1 \equiv s_2 + t_2 \pmod{n} \quad \text{and} \quad s_1 t_1 \equiv s_2 t_2 \pmod{n}.$$

Proof. Suppose $s_1 \equiv s_2 \pmod{n}$ and $t_1 \equiv t_2 \pmod{n}$. Then $n \mid (s_1 - s_2)$ and $n \mid (t_1 - t_2)$, so there exist integers k_1 and k_2 such that

$$s_1 - s_2 = nk_1 \quad \text{and} \quad t_1 - t_2 = nk_2.$$

Then

$$(s_1 + t_1) - (s_2 + t_2) = (s_1 - s_2) + (t_1 - t_2) = nk_1 + nk_2 = n(k_1 + k_2),$$

so $s_1 + t_1 \equiv s_2 + t_2 \pmod{n}$.

Similarly,

$$\begin{aligned} s_1 t_1 - s_2 t_2 &= s_1 t_1 - s_1 t_2 + s_1 t_2 - s_2 t_2 \\ &= s_1(t_1 - t_2) + (s_1 - s_2)t_2 \\ &= s_1(nk_2) + (nk_1)t_2 \\ &= n(s_1 k_2 + k_1 t_2), \end{aligned}$$

thus $s_1 t_1 \equiv s_2 t_2 \pmod{n}$. □

◆ $(\mathbb{Z}/n\mathbb{Z})^\times$: The subset of $\mathbb{Z}/n\mathbb{Z}$ consisting of all congruence classes with multiplicative inverses, i.e.,

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid \text{there exists } \bar{b} \in \mathbb{Z}/n\mathbb{Z} \text{ such that } \bar{a} \cdot \bar{b} = \bar{1}\}.$$

♣ Proposition 0.4 ► $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} \mid (a, n) = 1\}$

See Exercises 0.3.12 and 0.3.13.

1 Introduction to Groups

1.1 Basic Axioms and Examples

- ◆ **Binary Operation:** On a set G , it is the function $\star : G \times G \rightarrow G$. We write $a \star b$ for $\star(a, b)$, where $a, b \in G$.
- ◆ **Associative:** A binary operation $\star$ on a set G such that $a \star (b \star c) = (a \star b) \star c$ for all $a, b, c \in G$.
- ◆ **Commutative:** A binary operation $\star$ on a set G such that $a \star b = b \star a$ for all $a, b \in G$. If $a \star b = b \star a$ for some $a, b \in G$, we say that a and b **commute**.

✓ Example ► Binary Operations

- (1) $+$ and $\cdot$ are associative and commutative binary operations on $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$.
- (2) $-$ is not a binary operation on $\mathbb{Z}^+$ since $a - b \notin \mathbb{Z}^+$ when $a < b$, i.e., $- : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is not well-defined.
- (3) The operation of taking cross products in $\mathbb{R}^3$ is a non-associative and non-commutative binary operation on $\mathbb{R}^3$.

◆ **Closure:** Given a binary operation $\star$ on a set G with subset $H \subseteq G$, we say that H is **closed under** $\star$ if $a \star b \in H$ for all $a, b \in H$.

◆ **Group:** An ordered pair $(G, \star)$ for a set G and a binary operation $\star$ on G such that

- (1) $\star$ is associative, i.e., $a \star (b \star c) = (a \star b) \star c$ for all $a, b, c \in G$,
- (2) there exists an **identity element** $e \in G$ such that $e \star a = a \star e = a$ for all $a \in G$, and
- (3) for each $a \in G$, there exists $a^{-1} \in G$, called the **inverse** of a , such that $a \star a^{-1} = a^{-1} \star a = e$.

◆ **Abelian:** A group $(G, \star)$ is **Abelian** if $\star$ is commutative.

◆ **Finite Group:** A group $(G, \star)$ is **finite** if G is a finite set.

✓ Example ► Groups

- (1) $(G, +)$ is an Abelian group for $G = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ with $e = 0$ and $a^{-1} = -a$ for all $a \in G$.
- (2) The axioms of a vector space V imply that $(V, +)$ is an Abelian group for all $n \in \mathbb{Z}^+$.

◆ **Direct Product:** Given groups $(G, \star)$ and $(H, \diamond)$, their **direct product** is the group $(G \times H, *)$ where

$$G \times H = \{(g, h) \mid g \in G, h \in H\}$$

and $*$ is defined for all $(g_1, h_1), (g_2, h_2) \in G \times H$ as

$$(g_1, h_1) * (g_2, h_2) = (g_1 \star g_2, h_1 \diamond h_2).$$

♣ Proposition 1.1 ► Group Properties

Let $(G, \star)$ be a group.

- (1) The identity element $e \in G$ is unique.
- (2) For each $g \in G$, the inverse $g^{-1} \in G$ is unique.
- (3) $(g^{-1})^{-1} = g$ for all $g \in G$.
- (4) For all $g, h \in G$, $(g \star h)^{-1} = h^{-1} \star g^{-1}$.
- (5) For any $g_1, \dots, g_n \in G$, the value of $g_1 \star g_2 \star \dots \star g_n$ is independent of how the product is parenthesized, also known as the generalized associative law.

Proof.

- (1) Let $e_1, e_2 \in G$ be identity elements. Then

$$e_1 = e_1 \star e_2 = e_2.$$

(2) Let $g \in G$ and let $g_1^{-1}, g_2^{-1} \in G$ be inverses of g . Then

$$g_1^{-1} = g_1^{-1} \star e = g_1^{-1} \star (g \star g_2^{-1}) = (g_1^{-1} \star g) \star g_2^{-1} = e \star g_2^{-1} = g_2^{-1}.$$

(3) Note that $g^{-1}(g^{-1})^{-1} = e = g^{-1}g$. By the uniqueness of inverses, we have $(g^{-1})^{-1} = g$.

(4) Let $k = (g \star h)^{-1}$. Then

$$(g \star h) \star k = e.$$

Left multiply both sides by g^{-1} and use the associativity of $\star$ to obtain

$$\begin{aligned} g^{-1} \star ((g \star h) \star k) &= g^{-1} \star e \\ (g^{-1} \star g) \star (h \star k) &= g^{-1} \\ e \star (h \star k) &= g^{-1} \\ h \star k &= g^{-1}. \end{aligned}$$

Left multiply both sides by h^{-1} and simplify similarly to obtain

$$\begin{aligned} h^{-1} \star (h \star k) &= h^{-1} \star g^{-1} \\ (h^{-1} \star h) \star k &= h^{-1} \star g^{-1} \\ e \star k &= h^{-1} \star g^{-1} \\ k &= h^{-1} \star g^{-1}. \end{aligned}$$

(5) We proceed by induction on n . The base case $n = 1$ is trivial. For cases $n = 2$ and $n = 3$, the result follows from the associativity of $\star$. Suppose for all $k < n$ that the bracketing of a product of the k elements $h_1 \star \cdots \star h_k$ can be reduced to the form

$$h_1 \star (h_2 \star (\cdots \star (h_{k-1} \star h_k)) \cdots).$$

For notation, write the product above as $R(h_1, \dots, h_k)$. Consider a product of n elements $g_1 \star \cdots \star g_n$ with some arbitrary bracketing. Then we can write the product as

$$\underbrace{(g_1 \star \cdots \star g_i)}_{i \text{ factors}} \star \underbrace{(g_{i+1} \star \cdots \star g_n)}_{n-i \text{ factors}}$$

for some $1 \leq i < n$. By the inductive hypothesis, we may write the i factors as $R(g_1, \dots, g_i)$ and the $n - i$ factors as $R(g_{i+1}, \dots, g_n)$. Thus, we have

$$g_1 \star \cdots \star g_n = R(g_1, \dots, g_i) \star R(g_{i+1}, \dots, g_n).$$

To show that $g_1 \star \cdots \star g_n = R(g_1, \dots, g_n)$, we first show that $R(g_1, \dots, g_i) \star x = R(g_1, \dots, g_i, x)$ for any $x \in G$. We proceed by induction on i . The base case $i = 1$ is trivial. For $i = 2$, we have $R(g_1, g_2) \star x = (g_1 \star g_2) \star x = g_1 \star (g_2 \star x) = R(g_1, g_2, x)$ by the associativity of $\star$. Suppose for all $j < i$ that $R(g_1, \dots, g_j) \star x = R(g_1, \dots, g_j, x)$ for any $x \in G$. Noting that $R(g_1, \dots, g_i) = g_1 \star R(g_2, \dots, g_i)$, we have

$$\begin{aligned} R(g_1, \dots, g_i) \star x &= (g_1 \star R(g_2, \dots, g_i)) \star x \\ &= g_1 \star (R(g_2, \dots, g_i) \star x) \\ &= g_1 \star R(g_2, \dots, g_i, x) \\ &= R(g_1, \dots, g_i, x), \end{aligned}$$

where the second equality follows from the associativity of $\star$ and the third equality follows

from the inductive hypothesis.

Using the above claim and putting $x = R(g_{i+1}, \dots, g_n)$, we have

$$\begin{aligned} g_1 \star \dots \star g_n &= R(g_1, \dots, g_i) \star R(g_{i+1}, \dots, g_n) \\ &= R(g_1, \dots, g_i, R(g_{i+1}, \dots, g_n)) \\ &= R(g_1, \dots, g_n), \end{aligned}$$

where the last equality follows from the definition of R . Thus, by induction, the value of $g_1 \star \dots \star g_n$ is independent of how the product is parenthesized. $\square$

Note

We may drop the $\star$ symbol and write gh for $g \star h$ when the binary operation is clear from context. Moreover, we may write g^n for the product of n copies of $g \in G$, where $g^0 = e$ and $g^{-n} = (g^{-1})^n$ for all $n \in \mathbb{Z}^+$. Additionally, we may write 1 for the identity element $e \in G$.

Proposition 1.2 $\blacktriangleright$ Left and Right Cancellation Laws

Let G be a group with $g, h \in G$. Then the equations $gs = h$ and $tg = h$ have unique solutions $s, t \in G$, respectively. In particular,

- (1) if $gx = gy$, then $x = y$, and
- (2) if $xh = yh$, then $x = y$.

Proof. The equation $gs = h$ can be solved by left multiplying both sides by g^{-1} to obtain $s = g^{-1}h$, which is unique by the uniqueness of inverses. Similarly, $tg = h$ can be solved to obtain the unique solution $t = hg^{-1}$.

The first cancellation law follows from left multiplying both sides of $gx = gy$ by g^{-1} to obtain $x = y$, and the second cancellation law follows from right multiplying both sides of $xh = yh$ by h^{-1} to obtain $x = y$. $\square$

◆ **Order:** The smallest $n \in \mathbb{Z}^+$ for a group element $g \in G$ such that $g^n = 1$, denoted as $|g| = n$. If no such n exists, we say that g has **infinite order** and write $|g| = \infty$.

◆ **Multiplication/Group Table:** Put $G = \{g_1, g_2, \dots, g_n\}$ with $g_1 = 1$. Then the **multiplication table** or **group table** of G is the $n \times n$ matrix whose (i, j) th entry is $g_i g_j$ for all $1 \leq i, j \leq n$.